Comprehensive Portfolio Risk Analysis Report

Table of Contents

Detailed Interpretation and Impact Assessment

Executive Summary

This report provides an exhaustive risk analysis of a diversified Indian equity portfolio comprising five major stocks with equal weights. The analysis encompasses traditional risk metrics, advanced modeling techniques, and regulatory stress testing requirements. Each metric is thoroughly interpreted with practical implications for portfolio management and investment decision-making.

1. Returns and Volatility Analysis

Key Metrics:

- Mean Daily Return: 0.0604%
- Annualized Return: 15.22%
- Daily Volatility: 0.9012%
- Annualized Volatility: 14.31%

Interpretation and Impact:

The portfolio generates a robust annualized return of 15.22%, which is competitive for Indian equity markets. However, this comes with 14.31% annual volatility, indicating significant price fluctuations.

Practical Impact: For an investor with ₹1 crore, the expected annual gain is ₹15.22 lakh, but the portfolio value can fluctuate by approximately ±₹14.31 lakh annually. This volatility requires investors to have:

- Strong risk tolerance
- Long-term investment horizon (3-5 years minimum)
- Psychological preparedness for significant short-term losses

Portfolio Management Implications:

- Regular rebalancing may be needed to maintain target allocations

- Consider volatility targeting strategies during high-stress periods
- Diversification across asset classes could reduce overall portfolio volatility

2. Risk-Adjusted Performance (Sharpe Ratio)

Key Metrics:

- Sharpe Ratio: 1.064 (GOOD category)

Interpretation and Impact:

A Sharpe ratio of 1.064 indicates good risk-adjusted returns, meaning the portfolio earns 1.064 units of excess return for every unit of risk taken. This falls within the "acceptable" range (>1.0) but below the "strong" threshold (>2.0).

Benchmarking:

- Sharpe > 1.0: Acceptable
- Sharpe > 2.0: Strong
- Sharpe > 3.0: Exceptional

Investment Implications:

- The portfolio offers adequate compensation for risk taken
- Suitable for moderate to aggressive risk profiles
- Could be enhanced through better stock selection or tactical allocation adjustments
- Institutional investors would find this acceptable for equity allocations

3. Return Distribution Characteristics

Key Metrics:

- Skewness: 0.033 (near-symmetric)
- Excess Kurtosis: 0.146 (normal distribution)

Interpretation and Impact:

The portfolio exhibits near-symmetric return distribution with normal tail behavior. This is favorable because:

Positive Implications:

- Standard risk models (VaR, ES) are more reliable
- Lower probability of extreme tail events
- Predictable risk characteristics

Risk Management Impact:

- Normal distribution assumptions in risk models are justified
- Standard backtesting procedures are appropriate
- Lower need for exotic risk measures or extreme value theory

4. Maximum Drawdown Analysis

Key Metrics:

- Maximum Drawdown: -16.87%

Interpretation and Impact:

The worst peak-to-trough decline was 16.87%, which falls within acceptable ranges for equity portfolios (<20%).

Practical Impact for Investors:

- ₹1 crore invested at peak would decline to ₹83.13 lakh at the worst point
- Psychological stress during drawdown periods
- Potential liquidity constraints if forced to sell during drawdown

Risk Management Considerations:

- Investors need sufficient liquidity buffers
- Recovery time can be unpredictable
- May trigger stop-loss mechanisms or margin calls for leveraged positions
- Important for setting realistic expectations with clients

Mitigation Strategies:

- Dollar-cost averaging during drawdown periods
- Maintain emergency cash reserves
- Consider downside protection strategies (put options, structured products)

5. Value at Risk (VaR) and Expected Shortfall (ES)

Key Metrics:

- Daily VaR (95%): -1.38%
- Daily ES/CVaR (95%): -1.79%

Interpretation and Impact:

VaR Analysis:
On 95% of trading days, the portfolio will not lose more than 1.38%. Only 1 in 20 days will experience worse losses.

ES Analysis:
When tail events occur (worst 5% of days), the average loss is 1.79%. ES provides a more conservative estimate by capturing tail risk beyond VaR.

Practical Impact (₹1 Crore Portfolio):

- VaR: Maximum daily loss of ₹1.38 lakh on 95% of days
- ES: When bad days occur, average loss is ₹1.79 lakh

Regulatory Significance:

- Basel III requires banks to use ES (not just VaR) for capital calculations
- ES is a coherent risk measure, addressing VaR's mathematical limitations
- Critical for internal risk limits and regulatory reporting

Decision-Making Impact:

- Risk budgeting and position sizing
- Setting appropriate stop-loss levels
- Determining required capital buffers
- Client communication and expectation setting

6. VaR Backtesting Results

Key Metrics:

- VaR Violations: 131 out of 2,606 days
- Violation Rate: 5.03% vs. Expected 5.0%
- Test Result: PASS

Interpretation and Impact:

The model achieved excellent calibration with actual violations (5.03%) closely matching the theoretical expectation (5.0%).

Validation Significance:

- Model accuracy is confirmed through backtesting
- Risk estimates are neither too conservative nor too aggressive
- Regulatory compliance requirements are met

Ongoing Risk Management:

- Continuous backtesting required to ensure model remains valid
- If violations exceed expected rates, model recalibration needed
- Critical for maintaining regulatory approval and internal confidence

Action Framework:

- Violations < 3%: Model may be too conservative (overestimating risk)
- Violations > 7%: Model may be too aggressive (underestimating risk)
- Current 5.03%: Optimal calibration

7. Monte Carlo Simulation (10-Day Horizon)

Key Metrics:

- 10-Day MC VaR (95%): -4.09%
- 10-Day MC ES (95%): -5.32%

Interpretation and Impact:

Over a 10-day holding period, there's a 95% confidence that cumulative losses will not exceed 4.09%. When extreme scenarios occur, average losses are 5.32%.

Practical Impact (₹1 Crore Portfolio):

- Worst-case scenario (5% tail): Loss of ₹4.09 lakh over 10 days
- Average tail loss: ₹5.32 lakh over 10 days

Critical Applications:

1. **Liquidity Planning:** Essential for funds with redemption requirements

2. **Regulatory Capital:** Required for CCAR/DFAST stress testing
3. **Trading Desk Limits:** Multi-day risk limits for proprietary trading
4. **Client Reporting:** Realistic worst-case scenarios over typical holding periods

Advantages over Single-Day Models:

- Captures path dependency and compounding effects
- More realistic for actual investment horizons
- Better alignment with investor psychology and behavior

8. Regulatory Stress Testing

Key Metrics:

- 5-Sigma Stress Loss: -4.45%
- 10-Sigma Stress Loss: -8.95%
- Probability: ~1 in 1.7 million (5-sigma events)

Interpretation and Impact:

5-Sigma Event Analysis:
Despite extremely low probability (~1 in 1.7 million), such events have occurred:

- 1987 Black Monday crash
- 2008 Global Financial Crisis
- 2020 COVID-19 market crash

Practical Impact (₹1 Crore Portfolio):

- 5-sigma event: ₹4.45 lakh single-day loss
- 10-sigma event: ₹8.95 lakh single-day loss

Regulatory Context:

- SEBI/RBI/Basel III require adverse and severely adverse scenario testing
- Institutions must maintain capital to survive extreme but plausible events
- Stress testing results inform capital adequacy assessments

Risk Management Actions Required:

1. **Capital Reserves:** Maintain buffers for tail events
2. **Hedging Strategies:** Options, diversification, correlation reduction
3. **Liquidity Buffers:** Cash reserves for margin calls and redemptions
4. **Risk Limits:** Position size limits based on stress test outcomes
5. **Recovery Planning:** Predetermined actions during crisis scenarios

Conclusions and Recommendations

Overall Portfolio Assessment

The portfolio demonstrates:

- **Solid risk-adjusted returns** (Sharpe 1.064)
- **Well-calibrated risk models** (backtesting PASS)
- **Manageable downside risk** (drawdown within acceptable limits)
- **Normal distribution characteristics** (reducing model risk)

Key Risk Considerations

1. **Volatility Management:** 14.31% annual volatility requires active monitoring
2. **Tail Risk Preparedness:** Despite normal distribution, stress scenarios require preparation
3. **Liquidity Planning:** Multi-day VaR critical for redemption management
4. **Regulatory Compliance:** Ongoing backtesting and stress testing required

Strategic Recommendations

For Individual Investors:

- Maintain 6-month emergency fund separate from portfolio
- Consider dollar-cost averaging for new investments
- Set realistic expectations for volatility and drawdowns
- Review risk tolerance regularly, especially during stress periods

For Institutional Managers:

- Implement dynamic risk budgeting based on VaR/ES estimates
- Establish clear escalation procedures when risk limits are breached
- Consider tactical allocation adjustments during high-volatility periods
- Maintain comprehensive stress testing and scenario analysis programs

For Regulatory Compliance:

- Continue quarterly backtesting of risk models
- Update stress scenarios to reflect current market conditions
- Maintain capital buffers above minimum regulatory requirements
- Document risk management procedures and governance frameworks

This comprehensive analysis provides the foundation for informed risk management decisions and regulatory compliance while maintaining competitive returns in the Indian equity market.

Analysis Period: 10 Years (2,606 trading days)
Portfolio: Equal-weighted Indian equity (3M India, Adani Green, Axis Bank, Reliance, Bajaj Finance)
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